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Backstepping control based on neural network estimation

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ABSTRACT: This paper is implementing a backstepping controller to a quadrotor that is subject to unknown disturbances and uncertainty in dynamics. Estimation and approximation of unkowns and uncertainties are performed by using Radial Base Function Neural Network (RBFNN). Along with a backstepping controller, they provide robustness to the robotic systemdespite of the presence od uncertainties. The RBFNN's output layer is utilized as an estimator then compensation eliminates the undesired effect occurs due to uncertainties. As a result, faster error convergence is achievable. A Lyapunov function was used to analyze the closed loop system, and Matlab/Simulink was used to evaluate the system performance. The demonstrated results prove the efficiency of the proposed approach.

Keywords: Radial Base Function, neural network, Backstepping control, quadrotor, Unmanned Arial vehicles. Adaptive control system

1 INTRODUCTION

In the last decades, robotics technology has been developed remarkably sin terms of control systems and intelligence. Robot systems suffer from Nonlinearity, uncertain parameters, unmodeled and coupled dynamics in addition to external disturbances. Such challenges in robotic control systems are large areas of research. Many researchers have developed nonlinear controllers to overcome nonlinearity problems. For example, a terminal sliding mode is used to stabilize a control system as in [1], sliding mode control is utilized in many robotics applications [2], a sliding mode with second order is implemented [3], feedback linearization controllers were introduced in [4], super-twisting is used in [5,6]. Although the aforementioned controllers are successful, their accuracy depends on how accurate are the mathematical system model and system parameters. Additionally, changes in mass and inertia also reduce the efficiency of the controller's performance. To attain more robustness, researchers created adaptive controllers. To tackle unmodeled dynamics, inaccurate parameters, and external disturbances [7]. An adaptive backstepping controller is used [8], and a nonlinear time-varying systems was controlled by an adaptive control [9]. Adaptive controllers requires a predefined model structure which his not the case in most robots. To tackle this problem researchers designed observers as in [10,11]. An observer of three loops was designed to estimate and compensate unknowns [12]. Disturbance compensation is utilized in [13].

Radial Base Function Neural Networks (RBFNNs) are a different strategy for providing estimation of nonlinearity and uncertainty of robotic systems. The development of a controller incorporating neural network feedback was performed [14,15]. On the other hand, to improve the effectiveness of the controllers, the singularity issue was studied in many research projects. Super-Twisting Non-Singular Fast Terminal Control is built in [16].

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The backstepping control is robust for robotic structures with multiple coupled dynamics and unkowns [17,18]. Backstepping controllers provide a systematic framework for error tracking. The effectiveness of the backstepping depends on the exact modeling of the mechanical system.

A promossing solution to such problems is the use of Neural Networks (NNs) to estimate uncertain nonlinear functions where the neural network weights are tuned online without requiring a learning phase. However, further research on RBFNNs is required for the high computational cost and approximation lag.

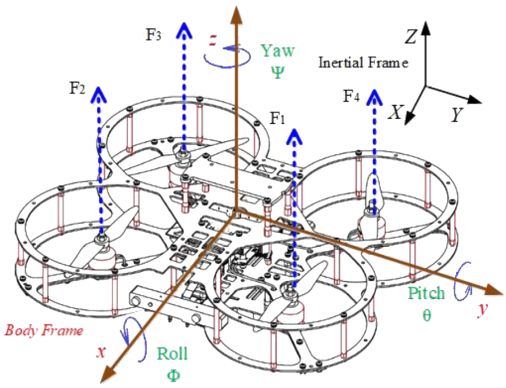

Figure 1. Quadrotor configuration.

In this article, the estimation ability of RBF Neural Networks and the effectiveness of backstepping control motivate us to implement such techniques to a quadrotor drone. In order to ensure the stability of the overall system, Lyapunov stability was examined. The RBFNN's response lag and high computational cost, both are eliminated by using the node number as a tuning variable, then selecting the minimum possible node number that achieves the desired stability. We used a quadrotor drone model to test the backstepping controller design with RBF approximation.

The main important highlight of the proposed design can be summarized as:

- *The RBF system provides an accurate estimatiom for imperfect modeling and varying parameters without the need for exact system modeling.*
- *The RBF system performs constant estimation, upper bound of model uncertainty is not necessary.*
- *The entire design of the Radial Base Functin and the Backstepping controller guarantees finite time error convergence and tracking.*

This paper is prepared as follows, introduction in Part I, the quadrotor model in Part II. The problem formulation and the design objective are in Part III, Section IV explains the RBFNN. Part V describes control design, Part VI shows simulation results. Part VII briefed the conclusion.

2 QUADROTOR DYNAMICS

The proposed controller is to be applied to a quadrotor drone. The quadrotor has four rotors, and variations in rotor velocities move the quadrotor in different directions. Opposite rotors direction is similar to cancel the gyroscopic effect. The quadrotor goes through many forces, moments, and gyroscopic effects, Figure 1, which have to be considered for developing the dynamic model [19]. Based upon Newton Euler formulas, the quadrotor model is given as [20,21]:

$$
\begin{aligned}
\ddot{x}_1 &= \ddot{\phi} = \dot{\theta}\dot{\psi}\frac{(i_{yy}-i_{zz})}{i_{xx}} - \dot{\theta}\omega_r\frac{j_r}{i_{xx}} - \frac{k_{fax}}{i_{xx}}\dot{\phi}^2 + \frac{1}{i_{xx}}u_2 \\
\ddot{x}_2 &= \ddot{\theta} = \dot{\phi}\dot{\psi}\frac{(i_{zz}-i_{xx})}{i_{yy}} + \dot{\phi}\omega_r\frac{j_r}{i_{yy}} - \frac{k_{fay}}{i_{yy}}\dot{\theta}^2 + \frac{1}{i_{yy}}u_3 \\
\ddot{x}_3 &= \ddot{\psi} = \dot{\theta}\dot{\phi}\frac{(i_{xx}-i_{yy})}{i_{zz}} - \frac{k_{faz}}{i_{zz}}\dot{\psi}^2 + \frac{1}{i_{zz}}u_4 \\
\ddot{x}_4 &= \ddot{x} = \frac{1}{m}(\sin\theta\cos\phi\cos\psi + \sin\phi\sin\psi)u_1 - \frac{k_{ftx}}{m}\dot{x} \\
\ddot{x}_5 &= \ddot{y} = \frac{1}{m}(\sin\theta\cos\phi\sin\psi - \sin\phi\cos\psi)u_1 - \frac{k_{fty}}{m}\dot{y} \\
\ddot{x}_6 &= \ddot{z} = \frac{-1}{m}(\cos\phi\cos\theta)u_1 + g - \frac{k_{ftz}}{m}\dot{z}
\end{aligned}
\tag{1}
$$

The variables are described as follows:

u_1, u_2, u_3 *and* u_4	: *Actual control input.*
u_x, u_y	: *Axillary control inputs:* $u_x = (\sin\theta\cos\phi\cos\psi + \sin\phi\sin\psi)$ $u_y = (\sin\theta\cos\phi\sin\psi - \sin\phi\cos\psi)$
x, y, z	: *The position in three coordinates respectively.*
ϕ, θ, *and* ψ	: *Rolling, pitching, and yawing angles.*
ϕ_d, θ_d *and* ψ_d	: *The desired angles.*
i_{xx}, i_{yy} *and* i_{zz}	: *Moments of inertia.*
$k_{fax}, k_{fay}, k_{faz}$	: *Aerodynamic friction coefficients.*
$k_{ftx}, k_{fty}, k_{ftz}$	: *The translation drag coefficients.*
m	: *Quadrotor mass.*
g	: *Gravity force.*
ω_i	: *Rotors' angular velocity* $(i = 1, 2, 3, 4)$.
ω_r	: *Angular equilibrium* $\omega_r = -\omega_1 + \omega_2 - \omega_3 + \omega_4$
d_c	: *Drag coefficient.*
B	: *Thrust coefficient.*
j_r	: *Rotor inertia.*
l_e	: *Length of the moment arm.*

We use axillary control inputs u_x, u_y to find the desired roll and pitch ϕ_d and θ_d which will be controlled by u_2 and u_3. Desired angles, roll and pitch [19,22]:

$$
\begin{aligned}
\phi_d &= \sin^{-1}\left(u_x\sin\psi_d - u_y\cos\psi_d\right) \\
\theta_d &= \sin^{-1}\left(\frac{u_x\cos\psi_d + u_y\sin\psi_d}{\cos\phi_d}\right)
\end{aligned}
\tag{2}
$$

3 PROBLEM FORMULATION

Considering the following nonlinear second-order robotic system described as:

$$
\begin{cases}
\dot{x}_1 = x_2 \\
\dot{x}_2 = f(x,t) + +g(x,t)u + d(t)
\end{cases}
\tag{3}
$$

Where x is the state variable, $f(\mathrm{x},\mathrm{t})$ and $g(x,t)$ are nonlinear and uncertain functions and $g(x,t) \neq 0$, $d(t)$ is the external disturbances. In real-time robotic systems, $f(x,t), g(x,t)$ and $d(t)$ cannot be precisely identified. Accurate approximation and estimation of the unknowns helps the controller to improve the entire performance.

3.1 *Objective*

– *Using the RBF-neural network to estimate uncertain dynamics and unknown disturbance to design a robust backstepping tracking control system.*

4 RADIAL BASIS FUNCTION NEURAL NETWORK

The Radial Base Function (RBF) is used to enhance control performance in various robot applications [23]. RBFNN can be used to learn the unknown upper bound of model uncertainty. They can also be used to estimate nonlinear continuous functions. Their objective is to provide estimation for uncertain modelling [14]. Three layers, Figure 2, make up the RBF: an input layer, an output layer, and a hidden layer. The input to the network is chosen as [24]:

$$\Gamma = \left[e^T \dot{e}^T\right]^T \tag{4}$$

The error and its derivative are denoted as $e = x_d - x$ and $\dot{e} = \dot{x}_d - \dot{x}$. Where, $x_d, \dot{x}_d$ are the desired trajectory and velocity respectively. $\phi\ (\Gamma(t))$ is the Gaussian radial basis function used for each node.

The hidden layer, is a linear combiner that maps nonlinearity into a new space. It performs a nonlinear transform of the input and output layer. RBF algorithm is given as [24]:

$$\begin{gathered} \varphi_i = \exp\left(-\frac{\|\Gamma - R_i\|^2}{b_i^2}\right) i = 1, 2, ..n \\ Q = \boldsymbol{W}^T \phi + \varepsilon \end{gathered} \tag{5}$$

Where,

$\boldsymbol{\phi} = [\varphi_1 \varphi_2 \ldots \varphi_n]^T$	*Gaussian function*
n	*Hidden layer number of nodes*
R_i	*ith node center vector*
b_i	*Smoothness control of interpolating function.*
E	*Approximation error*
Γ	*Input vector of neural network.*
Q	*Desired estimation*
$\boldsymbol{W}$	*Weight matrix*

The RBF neural network's approximation function is:

$$\widehat{Q}(x(t)) = \widehat{\boldsymbol{W}}^T \phi \tag{6}$$

Selecting an arbitrary positive initial value for the weight matrix $\widehat{\boldsymbol{W}}$. The network output can provide accurate approximation; thus, error tracking can converge to zero asymptotically.

The desired estimation of system (1) is the uncertain functions and the disturbance f, g, and d which can be approximated using the NNRBF as:

$$\begin{cases} f_\delta = \boldsymbol{W}_1^T \phi_1 + \varepsilon_1 \\ g = \boldsymbol{W}_2^T \phi_2 + \varepsilon_2 \end{cases} \tag{7}$$

Where $f_\delta = f + d$, ε_1 and ε_2 are approximation errors.

<u>*Assumption:*</u> approximation errors ε_1 and ε_2 are assumed to be bounded by a positive number ε_N, i.e. $\|\varepsilon_1 + \varepsilon_2\| \leq \varepsilon_N$.

The output estimation of the RBF-NN is given as:

$$\begin{cases} \tilde{f}_\delta = \tilde{\boldsymbol{W}}_1^T \phi_1 \\ \tilde{g} \ = \tilde{\boldsymbol{W}}_2^T \phi_2 \end{cases} \tag{8}$$

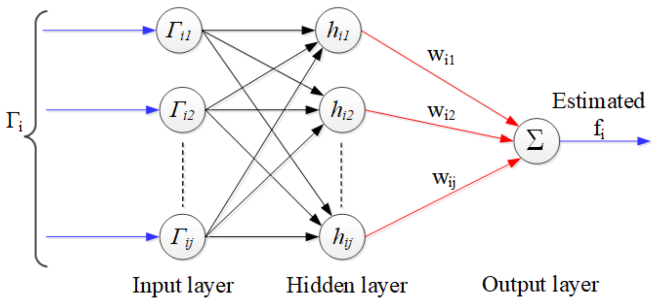

Figure 2. Neural network.

5 CONTROL DESIGN

Consider the nonlinear second-order robotic model described in (3). Backstepping controller design for the given system is achieved as described below. At first, we define the error and its derivative as e_1 and $\dot{e}_1$:

$$\begin{aligned} e_1 &= x_1 - x_{1d} \\ \dot{e}_1 &= \dot{x}_1 - \dot{x}_{1d} \end{aligned} \tag{9}$$

Where x_{1d} is the desired trajectory. To achieve $e_1 \to 0$, a Lyapunov function is sutilised as:

$$v_1 = \frac{1}{2} e_1^2 \tag{10}$$

Then,

$$\dot{v}_1 = e_1 \dot{e}_1 = e_1(\dot{x}_1 - \dot{x}_{1d}) = e_1(x_2 - \dot{x}_{1d}) \tag{11}$$

We choose $(x_2 - \dot{x}_{1d}) = -k_1 e_1$ where $k_1 > 0$, then we have:

$$\dot{v}_1 = -k_1 e_1^2 \tag{12}$$

The stabilization is obtained by introducing a virtual control input x_2 chosen as:

$$x_{2d} = \dot{x}_{1d} - k_1 e_1 \tag{13}$$

A second error is introduced for x_2 and x_{2d}:

$$e_2 = x_2 - x_{2d} \tag{14}$$

Then,

$$\dot{e}_2 = \dot{x}_2 - \dot{x}_{2d} = f(x,t) + g(x,t)u - \dot{x}_{2d} \tag{15}$$

To have a finite time errors convergence, a second Lyapunov analysis is required:

$$V_2 = V_1 + \frac{1}{2}e_2^2 = \frac{1}{2}\left(e_1^2 + e_2^2\right) \tag{16}$$

Then,

$$\begin{aligned} \dot{V}_2 &= e_1\dot{e}_1 + e_2\dot{e}_2 \\ e_1\dot{e}_1 &= e_1(\dot{x}_1 - \dot{x}_{1d}) = e_1(x2 - \dot{x}_{1d}) \\ &= e_1(x2d + e2 - \dot{x}_{1d}) = e_1(x2d + e2 - \dot{x}_{1d}) \\ &= -k_1e_1^2 + e_1e_2 \\ e_2\dot{e}_2 &= e_2(f(x,t) + g(x,t)u + d(t) - \dot{x}_{2d}) \end{aligned} \tag{17}$$

Then

$$\dot{V}_2 = -k_1e_1^2 + e_1e_2 + e_2(f(x,t) + g(x,t)u + d(t) - \dot{x}_{2d}) \tag{18}$$

If the nonlinear functions $f(x,t)$ and $g(x,t)$ are known exactly, it would be a routine nonlinear controller. The controller is designed to use the estimation provided by the RBF neural network, the control u is selected as:

$$u = \frac{1}{\widehat{g}}\left[-e_1 - \widehat{f}_\delta + \dot{x}_{2d} - k_2e_2\right] \tag{19}$$

Where, $f(x,t)$ and $g(x,t)$ are denoted as f and g respectively, $\widehat{f}_\delta$ is the estimation of $(f + d(t))$ and $\widehat{g}$ is the estimation of g. Substituting (19) in (18) we find,

$$\begin{aligned} \dot{V}_2 &= -k_1e_1^2 + e_1e_2 + e_2\left(\left(f - \widehat{f}_\delta\right)\right) + e_2\,(g - \widehat{g})u \\ \dot{V}_2 &= -k_1e_1^2 - k_2e_2^2 + e_2\tilde{f}_d + e_2\tilde{g}u \end{aligned} \tag{20}$$

Where $\tilde{f}_\delta = f + d(t) - \widehat{f}_\delta$ and $\tilde{g} = (g - \widehat{g})$ are the estimation errors in the dynamic system. Let the adaptive law as:

$$\dot{\widehat{Z}} = \Gamma h\xi^T - n\Gamma\|\eta\|\widehat{Z} \tag{21}$$

Where Z and $\tilde{Z}$ are given as:

$$Z = \begin{bmatrix} W_1 & 0 \\ 0 & W_2 \end{bmatrix}\tilde{Z} = \begin{bmatrix} \tilde{W}_1 & 0 \\ 0 & \tilde{W}_2 \end{bmatrix}, \tilde{Z} = Z - \widehat{Z}$$

And n is a positive number, Γ is a positive-definite matrix,

$$\Gamma = \begin{bmatrix} \Gamma_1 & 0 \\ 0 & \Gamma_2 \end{bmatrix} using\ \xi = [e_1e_2]^T \text{ and } V_2 = \frac{1}{2}\xi^T\xi.$$

Lyapunov stability for the backstepping control is selected as follows [25]:

$$V = \frac{1}{2}\boldsymbol{\xi}^T\boldsymbol{\xi} + \frac{1}{2}\mathrm{tr}\left(\tilde{\mathbf{Z}}^{\mathrm{T}}\mathrm{B}^{-1}\tilde{\mathbf{Z}}\right) \tag{22}$$

$$\dot{V} = \boldsymbol{\xi}^T\dot{\boldsymbol{\xi}} + \mathrm{tr}\left(\tilde{\mathbf{Z}}^{\mathrm{T}}\mathrm{B}^{-1}\dot{\tilde{\mathbf{Z}}}\right) \tag{23}$$

Using (7), (8), (18), and (23), it can be found that

$$\dot{V} = -k_1 e_1^2 - k_2 e_2^2 + e_2\left(\tilde{W}_1^T h_1 + \varepsilon_1\right) + e_2\left(\tilde{W}_2^T h_2 + \varepsilon_2\right) + \mathrm{tr}\left(\tilde{\mathbf{Z}}^{\mathrm{T}} \mathrm{B}^{-1} \dot{\tilde{\mathbf{Z}}}\right) \tag{24}$$

Let $\lambda_{\min}$ be the minimum eigenvalues of $K_e = [k_1 k_2]$. Substituting $\dot{\tilde{Z}} = -\dot{\hat{Z}}$ in (21) we find that $\dot{V} < 0$ if the following conditions are met:

$$\|\xi\| > \frac{\varepsilon_N + \frac{n}{4} Z_M^2}{\lambda_{\min}} \quad or \; \|\tilde{Z}\|_F > \frac{Z_M}{2} + \sqrt{\frac{Z_M^2}{4} + \frac{\varepsilon_N}{n}} \tag{25}$$

The proof of adaptive law design using RBF networks is further detailed in [26].

6 SIMULATION

In this part, the entire system is tested to verify the accuracy of trajectory tracking. Matlab/Simulink simulation is used for this purpose, parameter values are taken from a real quadrotor, and the trajectory is given as follows:

$$\begin{aligned} x_d &= \cos(t) \\ y_d &= \sin(t) \\ z_d &= 1 \\ \psi_d &= 0 \end{aligned} \tag{26}$$

The RBF estimation function (8) and the backstepping control (19) are used to create control and compensate for unknown dynamics and parameters in (3). Figure 3 displays the desired trajectory as given in (26) and the measured trajectory in three dimentions. Figures 4 and 5 displays the translational and orientational trajectories. The given graphs show the performance of the trajectory tracking system. The control's ability to provide precise estimation, bring the quadrotor to stability, and track the desired position is indicated by the fast convergence and the low values of errors. Figure 6 displays control signals. It is obvious that during the flight, the signals are continuous and never achieved saturation which indicate the capability to apply to a practical system to verify controller's quality and stability.

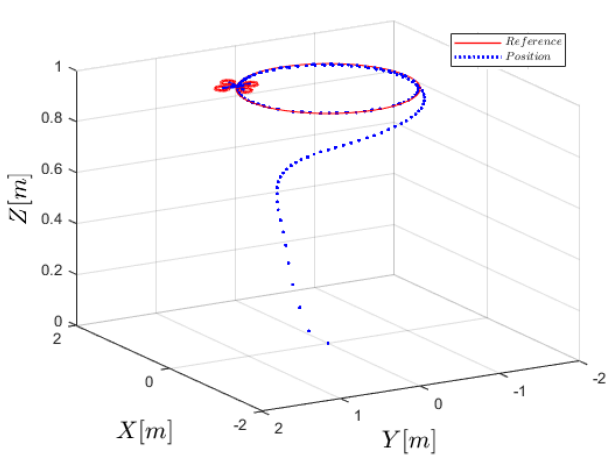

Figure 3. Trajectory tracking.

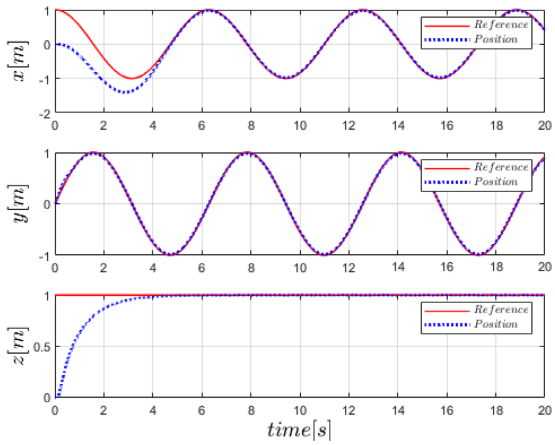

Figure 4. Trajectory tracking.

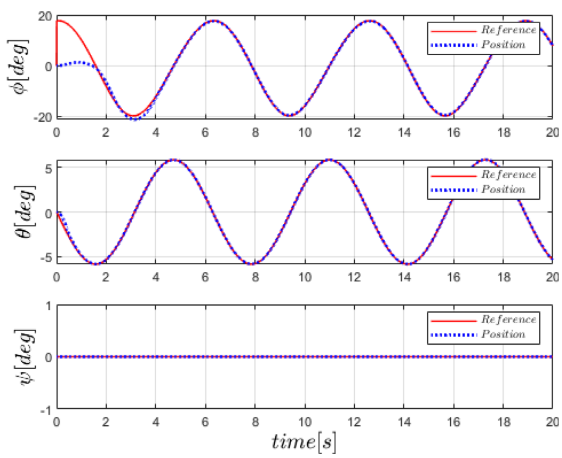

Figure 5. Response of roll, pitch, and yaw angles.

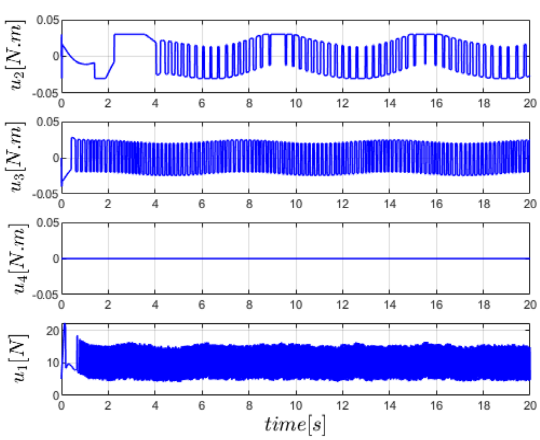

Figure 6. Controller signals.

7 CONCLUSION

This article proposed a design of a backstepping controller with a neural network radial base function, applied to a quadrotor. The coupled dynamics and the hard nonlinearities of the quadrotor make it ideal to test and prove the effective performance of a proposed system. Continuous estimation and compensation of uncertainties lead to quick convergence of errors. Performance and trajectory tracking are improved as a result. Matlab simulation and Lyapunov analysis demonstrated the system's effectiveness as well.

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